Tutorial 3

Ex. 8.5

Q42

$$\int \frac{\sin\theta}{\cos^2\theta+\cos\theta-2}\,d\theta$$

$u = \cos\theta$

$du = -\sin\theta\, d\theta$

$$= -\int \frac{1}{u^2+u-2}\,du$$

$$= -\int \frac{1}{(u-1)(u+2)}\,du$$

$$\frac{1}{(u-1)(u+2)} = \frac{A}{u-1} + \frac{B}{u+2}$$

Q46

$$\int \frac{1}{(x^{1/3}-1)\sqrt{x}}\,dx$$

Hint: $\boxed{x = u^6}$

$$= 2\int \frac{1}{\boxed{u^{2/3}}-1}\,du$$

$u = \sqrt{x} \Rightarrow x^{1/3} = u^{2/3}$

$du = \frac{1}{2\sqrt{x}}\,dx$

$dt = \frac{1}{3}t^{-2}\,du$

$du = 3t^2\,dt$

$t = u^{1/3}$

$dt = \frac{1}{3}u^{-2/3}\,du$

$= \frac{1}{3}(u^{1/3})^{-2}\,du$

$$= 2\int \frac{1}{t^2-1}(3t^2)\,dt$$

$$= 6\int \frac{t^{\boxed{2}}}{t^2-1}\,dt$$

$\boxed{\text{Long div.}} \rightarrow \boxed{\text{Partial Frac.}}$

$$\boxed{\frac{t^2}{t^2-1} = \frac{A}{t-1} + \frac{B}{t+1}} \quad (\text{Not True})$$

If $x^{m/n}$ appears, try

$$u = x^{1/n}$$

$$du = \frac{1}{n}x^{-\frac{(n-1)}{n}}\,dx$$

$$= \frac{1}{n}u^{-(n-1)}\,dx$$

Q48 $\displaystyle\int \frac{1}{x\sqrt{x+9}}\,dx$

$u = \sqrt{x+9}$

$u^2 = x+9$

$x = u^2 - 9$

$dx = 2u\,du$

$$= \int \frac{1}{(u^2-9)u}\,2u\,du$$

$$= \int \frac{2}{u^2-9}\,du \rightarrow \text{Partial Frac.}$$

$\displaystyle\int \frac{1}{x^{3/2}\sqrt{x+9}}\,dx$

$$= \int \frac{1}{\sqrt{x}\cdot x\sqrt{x+9}}\,dx$$

$u = \sqrt{x} \Rightarrow x = u^2$

$du = \dfrac{1}{2\sqrt{x}}$

$$= \square \int \frac{1}{u^2\sqrt{u^2+9}}\,du$$

$u = 3\tan\theta$

Q54

$$\int \frac{\boxed{\sqrt{x}}}{\boxed{\sqrt{2-\sqrt{x}}} + \boxed{\sqrt{x}}}\,\boxed{dx}$$

poly in u

$\boxed{dx} \to$ poly in u

$\sqrt{2-\sqrt{x}} \to u$

$\sqrt{x}\,(\ldots)$

Hard to do that

$u = \sqrt{2-\sqrt{x}}$

$u^2 = 2 - \sqrt{x}$

$\sqrt{x} = 2 - u^2 \to$ poly. in u

$x = (2-u^2)^2$

$dx = (\text{polynomial in } u)\,du$

Q50 $\displaystyle\int \frac{1}{x^6(x^5+4)}\,dx$

(Q49)

$\displaystyle\int \frac{1}{x(x^4+1)}\,dx$

Hint: Mult. by $\dfrac{x^3}{x^3}$

$$\int \frac{1}{x^m(x^n+k)}\,dx$$

If $\underline{m+n-1}$ is a multiple of $\underline{n}$, try $u=\underline{x^n}$.

$$\int \frac{\boxed{x^{n-1}}}{\underbrace{x^{m+n-1}}_{x^{\ell n}\,\to\,u^{\ell}}\underbrace{(x^n+k)}_{u}}\,dx$$

cancelled

Ex. 8.8

<u>Q44</u> $\displaystyle\int_0^1 \frac{e^{-\sqrt{x}}}{\sqrt{x}}\,dx$

$$= \lim_{b\to 0^+} \int_b^1 \frac{e^{-\sqrt{x}}}{\sqrt{x}}\,dx$$

$u = \sqrt{x}$ When $x=1$, $u=1$

$du = \dfrac{1}{2\sqrt{x}}\,dx$ When $x=b$, $u=\sqrt{b}$

$$\int_b^1 \frac{e^{-\sqrt{x}}}{\sqrt{x}}\,dx = 2\int_{\sqrt{b}}^1 e^{-u}\,du$$

$$= \underline{2\left[-e^{-u}\right]_{\sqrt{b}}^1} \rightarrow \text{Not } 0$$

$$= \lim_{b\to 0^+} \underline{2\left[-e^{-u}\right]_{\sqrt{b}}^1}$$

Q54 $\displaystyle\int_0^\infty \frac{d\theta}{1+e^\theta}$

$0 < e^\theta \leq 1+e^\theta$ for $\theta \in (0,\infty)$

$\dfrac{1}{1+e^\theta} \leq \dfrac{1}{e^\theta}$ for $\theta \in (0,\infty)$

$$\int_0^\infty \frac{1}{e^\theta}\,d\theta = \lim_{b\to\infty}\int_0^b \frac{1}{e^\theta}\,d\theta$$

$$= \lim_{b\to\infty}\left[-e^{-\theta}\right]_0^b$$

$$= \lim_{b\to\infty}\left[-e^{-b}+1\right]$$

$$= 1$$

$\therefore \displaystyle\int_0^\infty \frac{1}{e^\theta}\,d\theta$ converges.

$\therefore \displaystyle\int_0^\infty \frac{1}{1+e^\theta}\,d\theta$ converges by Direct Comparison Test.

"Hierarchy"

$(p>0)$

$$\ln x \ll x^p \ll e^x$$

$$\lim_{x\to\infty} \frac{\ln x}{x^p} = 0$$

$$\lim_{x\to\infty} \frac{x^p}{e^x} = 0$$

$$\int_a^\infty \frac{1}{e^x}\,dx \ll \int_a^\infty \frac{1}{x^p}\,dx \ll \int_a^\infty \frac{1}{x^q}\,dx$$

(Converge) $\quad$ ~~$0<p$~~ $(p>1)$ (Converge) $\quad$ $0<q\leq 1$ (Diverge)

$$\ll \int_a^\infty \frac{1}{\ln x}\,dx \rightarrow \underline{\underline{Q62}}$$

(Diverge)

Q62 $\displaystyle\int_2^\infty \frac{1}{\ln x}\,dx$

$$\boxed{\frac{1}{x} < \frac{1}{\ln x}}$$

Note that

$0 < \ln x < x$ for $x \geq 2$

$$\frac{1}{x} < \frac{1}{\ln x}$$

$\displaystyle\int_2^\infty \frac{1}{x}\,dx$ diverges.

$\therefore \displaystyle\int_2^\infty \frac{1}{\ln x}\,dx$ diverges by Direct comparison Test.

Q58 $\displaystyle\int_2^\infty \frac{x}{\sqrt{x^4-1}}\,dx$

$$\frac{x}{\sqrt{x^4-1}} \sim \frac{x}{\sqrt{x^4}} \sim \frac{x}{x^2} \sim \boxed{\frac{1}{x}}$$

Let $f(x) = \dfrac{x}{\sqrt{x^4-1}}$

$g(x) = \dfrac{1}{x}$

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \cdots = 1 > 0$$

Limit Comparison Test

Since $\displaystyle\int_2^\infty \frac{1}{x}\,dx$ diverges, $\displaystyle\int_2^\infty \frac{x}{\sqrt{x^4-1}}\,dx$ diverges by